



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECJ2203: Software Engineering

Project Proposal

Web-Based System for KADA's Cooperative

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Faculty of Computing

Prepared by: AlphaTech

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1. Introduction

1.1 The Goal

The Kemubu Agricultural Development Board (KADA) is an institution within the Ministry of Agriculture and Food Security that promotes and encourages the economic and social growth of farmers in the Kemubu region. Within KADA there is an organization called KADA's staff cooperative whose members unite voluntarily to achieve common interests in the economic, social and cultural fields. The cooperative only allows employees that work under KADA to join as a member of the cooperative. The cooperative offers benefits such as loan services for business purposes, emergencies and welfare funds for its members.

Currently, KADA's staff cooperative runs most of its processes manually from applying to become a member of the cooperative, applying for loans, report generation and data storage in a form of a hard copy facilitated by a clerk. An example of a problem from this is that when applying for a loan using an application form is that it is prone to inconsistency in data entries from manual data entries. These inconsistencies can cause significant errors, as data entered manually is prone to human error. It is also inefficient to always fill out repetitive information such as employees' names and ID numbers for applications. Other problems include data management issues, limitations in data collection and analysis and reliance on human resources. These problems raise a need for a system that automates many of the processes within the cooperative to reduce manual workload and minimizing errors, thus enhancing overall operational efficiency.

The main goal of the project is to propose a system that will be a website as an alternative to the manual system currently used by KADA. The system will still retain some functionality but with additional modifications that increase efficiency to enhance the working experience of the entities involved. Hence, the aim of this project is to develop a web-based system with user friendly interface and fulfills user requirements.

The objectives of this project include:

- i. To design a web-based system that fulfills user requirements.
- ii. To develop a complete and ready-to-use web system.
- iii. Run and test the developed system to ensure the user's needs are met.

1.2 The Scope

The scope of this project is to propose a system that will be a website that replaces manual systems by using multiple technologies to ensure the smooth-running system for KADA. Each of this technology has its own functionality to help the development of this system. The system will have some existing functionality but with additional changes using these technologies will increase efficiency to enhance the working experience because the technologies used will develop a web server with a relational database management system. The system will cover functionalities such as user authentication, membership application, membership application, administrative processes and successful payment notification. The system will not cover scopes that utilizes payment gateways as the main feature such as making a payment through the system, buying share modal and claiming of member benefits.

These are the technology that will be applied in the system:

- i. Apache HTTP Server
- ii. PHP Technology
- iii. MySQL
- iv. JavaScript
- v. CSS
- vi. HTML
- vii. Bootstrap
- viii. API

Table 1.1 shows the technology and its description to be used to develop this system project.

No.	Technology	Description
1.	Apache HTTP Server	Used to handle high volumes of HTTP requests, managing client connections, and serving web content efficiently.
2.	PHP Technology	Will be used for dynamic web pages, form submissions, and database interaction
3.	MySQL	Utilizing PHP scripts to retrieve, update, and manage information.
4.	JavaScript	Create dynamic and interactive web pages, faster load content to reduce response time.
5.	CSS	It will be used to design the interface to improve user experience.
6.	HTML	The backbone of the website that defines the content and structure of web content.
7.	Bootstrap	This will be used to determine the design of the website from typography, forms, buttons, navigation, and other interface components.
8.	API	Allows the use of other applications within the website.

Table 1.1 shows the technology and its description to be used to develop this system project.

2. Software Process Model

2.1 The Goal

A software process model is a collection of interconnected steps that result in software development. These efforts could entail developing software from the ground up or modifying an existing system. The purpose of a software process model is to provide guidance for directing and coordinating tasks so that the final output and objectives are met as efficiently as possible.

There are several process models available to meet different standards. A software process model defines the following:

- Determine what the system should do
- Organization of the system and system implementation
- Tasks that needs to be performed
- The input and output of each task
- The preconditions and postconditions for each task
- The sequence and flow of this task

2.2 The Chosen Software Process Model

The software process model that we used as the approach in this proposed system development on KADA system is the Waterfall model. The reason for our choice is that the Waterfall model also offers many advantages in the Software Development Life Cycle(SDLC). It also ensures a clear documentation and well defined phases making it more easier to manage and to track progress within the team. When compared to other methodologies, Waterfall is pretty straightforward and more focused for each phase. Waterfall model relies on teams following a sequence of steps and never moving forward until the previous phase has been completed. In the first phase, the requirements and quality standards. The requirement gathering was collected through an online meeting with a representative of KADA. After the user requirements are collected, the organization of the system is designed based on the user requirements. Next is the implementation phase where

the technical parts of the website are implemented using the technologies stated in *Table 1.1*. After that, the testing phase, this phase will have 4 components that need to be done which are unit testing, integrated testing, system testing and acceptance testing. These components will need to be done sequentially as listed. Unfortunately, this project will only cover until the testing phase.

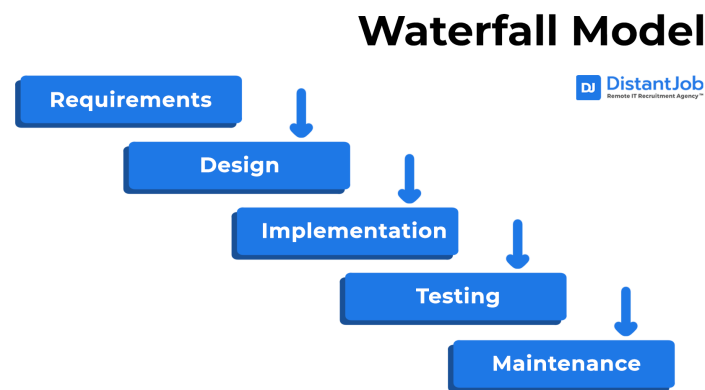


Figure 2.1 Waterfall Methodology

2.2 The Chosen Software Process Model

The project aims to include all the information and activities that are required to develop KADA's web-based system. Figure 2.2 shows five major activities which are Requirement Analysis Phase, Software Design Phase, Software Implementation Phase & Testing Phase . The assumption of the project is to be completed in about 13 weeks

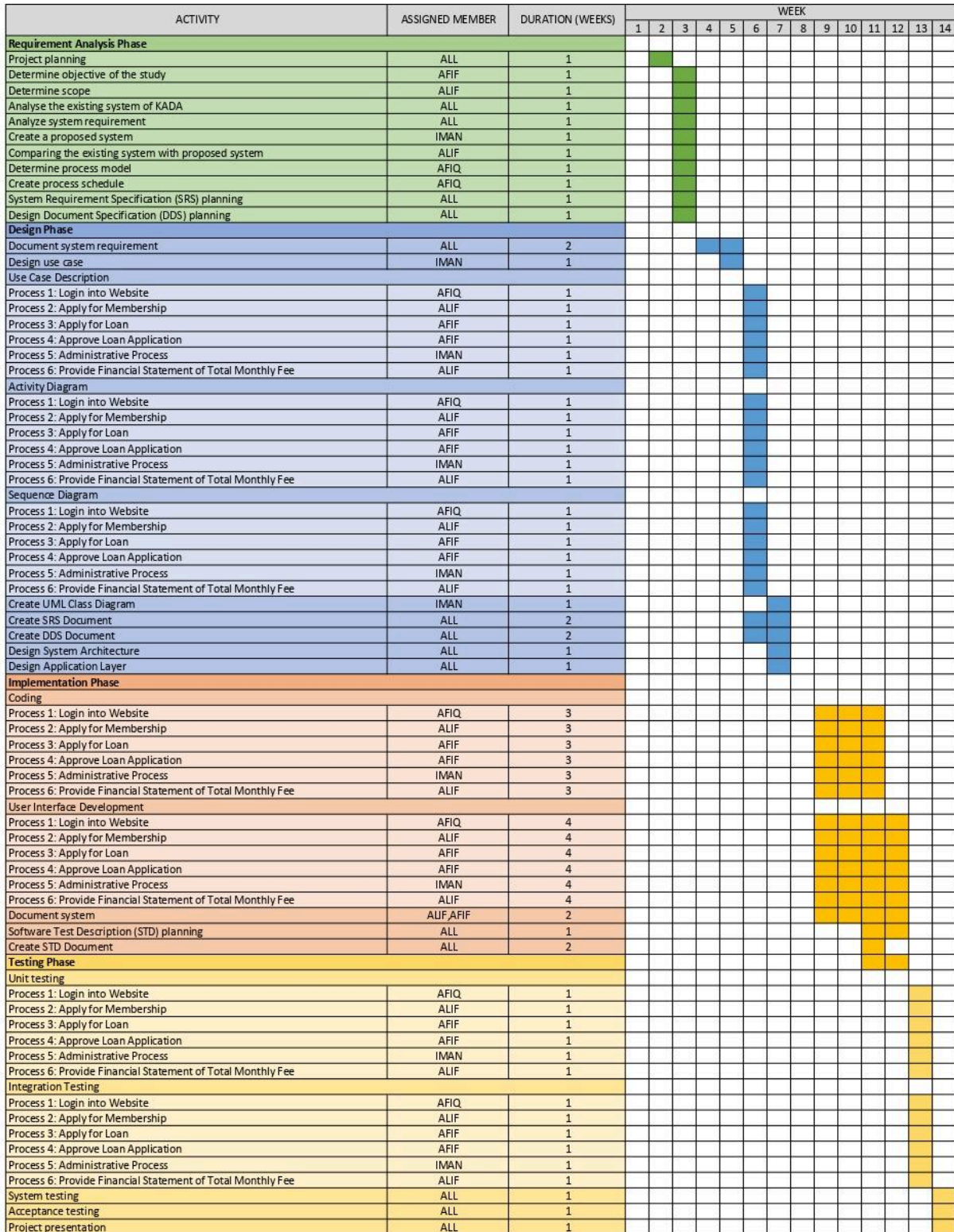


Figure 2.2 Gantt Chart

The project will begin in the first week with requirement analysis, which entails gathering requirements in order to analyze KADA's Cooperative's present manual system. This stage's deliverable is the Systems Requirements Specification. The project will then go to software design, which will take around four weeks. It entails designing the system interface as well as the architectural design. After a week, implementation will begin, which will take around 5 weeks. During this phase, the design of KADA's web-based system will be turned into a functioning prototype to test the functionality of the new features. Unit testing entails assessing the smallest testable components or aspects of the system separately and independently to ensure appropriate operation. The system testing will next proceed, with the KADA's web-based system being assessed as a whole and its compliance with its set requirements evaluated. A System Testing Document will be given to help with the testing process. Finally, at the end of the project, the system will be delivered to the KADA cooperative for acceptance testing.